

Multiple Choice (10 marks)

1. What particle always has a mass of one atomic mass unit (amu) and no charge?
 - (a) a neutron
 - (b) a proton
 - (c) an electron
 - (d) an atom
2. Which sentence could not explain why the first place swimmer was faster than the second place swimmer?
 - (a) The winner experienced less water resistance.
 - (b) There was more gravitational pull on the winner.
 - (c) Friction slowed down the second place swimmer.
 - (d) There was more force in the winner's kicks and strokes.
3. The chemical symbol for niobium is Nb. Based on its symbol, what can be concluded about niobium?
 - (a) Niobium is a compound.
 - (b) Niobium is an element.
 - (c) Niobium is a metal.
 - (d) Niobium is a mixture.
4. How many times does Earth rotate on its axis in one day?
 - (a) once
 - (b) twice
 - (c) 24 times
 - (d) 365 times
5. Which organ in a frog has a function similar to the function of lungs in a bird?
 - (a) kidney
 - (b) skin
 - (c) liver
 - (d) heart

True / False (10 marks)

Answer True or False

6. True or False: The handle of a frying pan is often coated in rubber because rubber offers a nonstick surface.
7. True or False: A paramecium can conjugate, allowing it to exchange genetic material with another paramecium.
8. True or False: Both burning coal and exploding fireworks are chemical reactions that release energy.
9. True or False: The best way to organize data on plant growth is by date, plant height, and soil type to identify patterns effectively.
10. True or False: Snow is a form of water that commonly appears when the temperature is below freezing.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the labeling of an object's motion can change while its velocity remains constant, providing examples to illustrate your points.
12. Calculate the momentum of an 8.0 kg bowling ball rolling at 2.0 m/s. Discuss how mass and velocity influence momentum and its implications in real-world scenarios.

End of Paper